

Homework 1

Wesley Park Linear Algebra Sep 16, 2026

Problem 1

(a)

Final answer: 24

(b)

$$v + w = [1, 4, -2]^T + [-4, 2, 6]^T = [-3, 6, 4]^T$$

$$\|v + w\| = \sqrt{(-3)^2 + 6^2 + 4^2} = \sqrt{9 + 36 + 16} = \sqrt{61} \approx 7.81$$

[ANSWER: $\sqrt{61} \approx 7.81$]

(c)

The angle satisfies $\cos \theta = (u \cdot w) / (\|u\| \|w\|)$.

$$u \cdot w = (2)(-4) + (-1)(2) + (3)(6) = -8 - 2 + 18 = 8$$

$$\|u\| = \sqrt{4 + 1 + 9} = \sqrt{14}, \quad \|w\| = \sqrt{16 + 4 + 36} = \sqrt{56} = 2\sqrt{14}$$

$$\cos \theta = 8 / (\sqrt{14} \cdot 2\sqrt{14}) = 8 / 28 = 2/7$$

$$\theta = \arccos(2/7) = 73.398\dots^\circ$$

[ANSWER: $\theta \approx 73.40^\circ$]

Problem 2

(a)

AB is undefined: A is 3×3 and B is 2×3 , so the 3 columns of A do not match the 2 rows of B . BA is fine (2×3 times 3×3), and it comes out 2×3 :

$$\text{row 1} = [1(2) + 0(1) - 2(0), 1(-1) + 0(3) - 2(4), 1(0) + 0(-2) - 2(1)] = [2, -9, -2]$$

$$\text{row 2} = [3(2) + 1(1) + 1(0), 3(-1) + 1(3) + 1(4), 3(0) + 1(-2) + 1(1)] = [7, 4, -1]$$

$$[\text{ANSWER: } BA = [2 \ -9 \ -2]$$

$$[7 \ 4 \ -1]]$$

(b)

Av is 3×3 times $3 \times 1 = 3 \times 1$:

$$Av = [2(1) - 1(-2) + 0(3), 1(1) + 3(-2) - 2(3), 0(1) + 4(-2) + 1(3)]^T = [4, -11, -5]^T$$

VA is undefined: v is 3×1 , so it has 1 column, but A has 3 rows, and $1 \neq 3$.

$$[\text{ANSWER: } Av = [4, -11, -5]^T, \ VA \text{ undefined}]$$

(c)

Aw is undefined: A has 3 columns and w (1×3) has only 1 row.

WA is 1×3 times $3 \times 3 = 1 \times 3$:

$$WA = [2(2) + 0(1) - 1(0), 2(-1) + 0(3) - 1(4), 2(0) + 0(-2) - 1(1)] = [4, -6, -1]$$

$$[\text{ANSWER: } Aw \text{ undefined, } WA = [4 \ -6 \ -1]]$$

(d)

vw is 3×1 times 1×3 , so it is a 3×3 outer product:

$$vw = [1(2) \ 1(0) \ 1(-1)] \ [2 \ 0 \ -1]$$

$$[-2(2) \ -2(0) \ -2(-1)] = [-4 \ 0 \ 2]$$

$$[3(2) \ 3(0) \ 3(-1)] \ [6 \ 0 \ -3]$$

wv is 1×3 times 3×1 , so it is 1×1 : $wv = 2(1) + 0(-2) + (-1)(3) = -1$

$$[\text{ANSWER: } vw \text{ above, } wv = [-1]]$$

(e)

A is square, so $A^2 = AA$ is defined:

$$\text{row 1} = [2(2)-1(1)+0(0), 2(-1)-1(3)+0(4), 2(0)-1(-2)+0(1)] = [3, -5, 2]$$

$$\text{row 2} = [1(2)+3(1)-2(0), 1(-1)+3(3)-2(4), 1(0)+3(-2)-2(1)] = [5, 0, -8]$$

$$\text{row 3} = [0(2)+4(1)+1(0), 0(-1)+4(3)+1(4), 0(0)+4(-2)+1(1)] = [4, 16, -7]$$

$$[\text{ANSWER: } A^2 = \begin{bmatrix} 3 & -5 & 2 \\ 5 & 0 & -8 \\ 4 & 16 & -7 \end{bmatrix}]$$

$$\begin{bmatrix} 5 & 0 & -8 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 16 & -7 \end{bmatrix}]$$

(f)

Square each entry of B:

$$B^2 = [1^2 \ 0^2 \ (-2)^2] \ [1 \ 0 \ 4]$$

$$[3^2 \ 1^2 \ 1^2] = [9 \ 1 \ 1]$$

$$[\text{ANSWER: } B^2 = \begin{bmatrix} 1 & 0 & 4 \\ 9 & 1 & 1 \end{bmatrix}]$$

Problem 3

(a)

Augmented matrix, then eliminate below the first pivot:

$$\begin{bmatrix} 1 & 2 & -1 & | & -1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 3 & 1 & | & 2 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 1 & 2 & | & -1 \end{bmatrix}$$

$$R2 \rightarrow R2 - 2R1 \quad R3 \rightarrow R3 + R1 \quad R2 \rightarrow -R2$$

$$\begin{bmatrix} 1 & 2 & -1 & | & -1 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 & -1 & | & -1 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 & -1 & | & -1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -1 & 3 & | & 4 \end{bmatrix} \quad \begin{bmatrix} 0 & -1 & 3 & | & 4 \end{bmatrix} \quad \begin{bmatrix} 0 & 1 & -3 & | & -4 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 1 & 2 & | & -1 \end{bmatrix} \quad \begin{bmatrix} 0 & 3 & 1 & | & -2 \end{bmatrix} \quad \begin{bmatrix} 0 & 3 & 1 & | & -2 \end{bmatrix}$$

$$R3 \rightarrow R3 - 3R2 \quad R3 \rightarrow (1/10)R3$$

$$\begin{bmatrix} 1 & 2 & -1 & | & -1 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 & -1 & | & -1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & -3 & | & -4 \end{bmatrix} \quad \begin{bmatrix} 0 & 1 & -3 & | & -4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 10 & | & 10 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 & 1 & | & 1 \end{bmatrix}$$

Back substitution: $x_3 = 1$, then $x_2 = -4 + 3(1) = -1$, then $x_1 = -1 - 2(-1) + 1 = 2$.

There are three pivots and no free variables, so the solution is unique.

[ANSWER: $x = [2, -1, 1]^T$]

(b)

$$[1 \ 2 \ -1 \mid 4]$$

$$[2 \ 5 \ 1 \mid 6]$$

$$[3 \ 7 \ 0 \mid 10]$$

$$R_2 \rightarrow R_2 - 2R_1$$

$$R_3 \rightarrow R_3 - 3R_1$$

$$R_3 \rightarrow R_3 - R_2$$

$$[1 \ 2 \ -1 \mid 4] \quad [1 \ 2 \ -1 \mid 4] \quad [1 \ 2 \ -1 \mid 4]$$

$$[0 \ 1 \ 3 \mid -2] \quad [0 \ 1 \ 3 \mid -2] \quad [0 \ 1 \ 3 \mid -2]$$

$$[3 \ 7 \ 0 \mid 10] \quad [0 \ 1 \ 3 \mid -2] \quad [0 \ 0 \ 0 \mid 0]$$

$$R_1 \rightarrow R_1 - 2R_2$$

$$[1 \ 0 \ -7 \mid 8]$$

$$[0 \ 1 \ 3 \mid -2]$$

$$[0 \ 0 \ 0 \mid 0]$$

Column 3 has no pivot, so z is free: set $z = t$. Then $x = 8 + 7t$ and $y = -2 - 3t$, and the system has infinitely many solutions.

[ANSWER: $[x, y, z]^T = [8, -2, 0]^T + t[7, -3, 1]^T, t \in \mathbb{R}$]

Problem 4

(a)

Let x_1 = mass of Alloy 1 used (kg), x_2 = mass of Alloy 2 used (kg), and x_3 = mass of Alloy 3 used (kg). The 100 kg product contains 36 kg copper, 44 kg zinc, and 20 kg nickel, and no metal is lost, so for each metal the mass in equals the mass out:

$$\text{copper: } 0.5x_1 + 0.2x_2 + 0.4x_3 = 36$$

$$\text{zinc: } 0.3x_1 + 0.7x_2 + 0.2x_3 = 44$$

$$\text{nickel: } 0.2x_1 + 0.1x_2 + 0.4x_3 = 20$$

(b)

Multiply all three rows by 10 so the entries are integers:

$$[5 \ 2 \ 4 \mid 360]$$

$$[3 \ 7 \ 2 \mid 440]$$

$$[2 \ 1 \ 4 \mid 200]$$

(c)

$$R_1 \rightarrow R_1 - 2R_3 \quad R_2 \rightarrow R_2 - 3R_1 \quad R_3 \rightarrow R_3 - 2R_1$$

$$[1 \ 0 \ -4 \mid -40] \quad [1 \ 0 \ -4 \mid -40] \quad [1 \ 0 \ -4 \mid -40]$$

$$[3 \ 7 \ 2 \mid 440] \quad [0 \ 7 \ 14 \mid 560] \quad [0 \ 7 \ 14 \mid 560]$$

$$[2 \ 1 \ 4 \mid 200] \quad [2 \ 1 \ 4 \mid 200] \quad [0 \ 1 \ 12 \mid 280]$$

$$R_2 \rightarrow (1/7)R_2 \quad R_3 \rightarrow R_3 - R_2 \quad R_3 \rightarrow (1/10)R_3$$

$$[1 \ 0 \ -4 \mid -40] \quad [1 \ 0 \ -4 \mid -40] \quad [1 \ 0 \ -4 \mid -40]$$

$$[0 \ 1 \ 2 \mid 80] \quad [0 \ 1 \ 2 \mid 80] \quad [0 \ 1 \ 2 \mid 80]$$

$$[0 \ 1 \ 12 \mid 280] \quad [0 \ 0 \ 10 \mid 200] \quad [0 \ 0 \ 1 \mid 20]$$

Back substitution: $x_3 = 20$, $x_2 = 80 - 2(20) = 40$, $x_1 = -40 + 4(20) = 40$.

Check: total $40 + 40 + 20 = 100$ kg, copper $20 + 8 + 8 = 36$ kg, so it works.

[ANSWER: 40 kg of Alloy 1, 40 kg of Alloy 2, 20 kg of Alloy 3]

Problem 5

(a)

$$R_2 \rightarrow R_2 - 2R_1 \quad R_3 \rightarrow R_3 - 3R_1 \quad R_3 \rightarrow R_3 - R_2 \quad R_1 \rightarrow R_1 - R_2$$

$$\begin{bmatrix} 1 & 2 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 6 & 4 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

$$[\text{ANSWER: RREF}(A) = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}]$$

(b)

$\text{rank}(A) = 2$, since the RREF has two pivot positions (columns 1 and 3), which is the same as saying it has two nonzero rows.

(c)

0 solutions: impossible, because A is not the zero matrix, so the rows can never all wipe out and leave a contradiction.

Exactly 1 solution: impossible. Column 2 has no pivot, so x_2 is a free variable in every consistent case, and $\text{rank}(A) = 2 < 3$ unknowns means uniqueness can't happen.

Infinitely many solutions: possible. Take $b = [1, 3, 4]^T$ (or $b = 0$): the system is consistent, and the free variable x_2 gives a whole line of solutions.

Problem 6

(a)

Row reduce the matrix whose columns are v_1, v_2, v_3 :

$$\begin{array}{l} \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 3 & -1 \end{bmatrix} \quad R_3 \rightarrow R_3 - 2R_1 \quad R_3 \rightarrow R_3 + R_2 \\ \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & -1 & -1 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \\ \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \end{array}$$

Only two pivots for three vectors, so the set is linearly dependent.

The dependence is $v_3 = v_2 - 2v_1$, since $[2, 1, 3]^T - [2, 0, 4]^T = [0, 1, -1]^T$.
 v_1 and v_2 are independent of each other, so together they sweep out a plane through the origin, and v_3 adds nothing new.

[ANSWER: dependent; span is a plane through the origin in $\mathbb{R}^3$]

(b)

Three vectors in $\mathbb{R}^3$, so I can just take the determinant of the matrix with those columns and see if it is zero:

$$\begin{vmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{vmatrix}$$

Cofactor expansion along the first row:

$$\det = 1(1 \cdot 1 - 0 \cdot 1) - 0(1 \cdot 1 - 0 \cdot 0) + 1(1 \cdot 1 - 1 \cdot 0) = 1(1) - 0 + 1(1) = 2$$

$\det = 2 \neq 0$, so the only solution of $c_1 v_1 + c_2 v_2 + c_3 v_3 = 0$ is the trivial one and the vectors are linearly independent. Three independent vectors in $\mathbb{R}^3$ span $\mathbb{R}^3$.

[ANSWER: independent; span is all of $\mathbb{R}^3$]